

Individuals with cardiometabolic risk factors are at higher risk for early-onset NAFLD

To the Editor:

Early-onset of non-alcoholic fatty liver disease (NAFLD) is linked to worse outcomes, and individuals under 45 years old

with NAFLD have a significantly higher risk of cancer compared to other age groups.¹ One-third of chronic liver disease-related deaths occur in patients with NAFLD under the age of 30

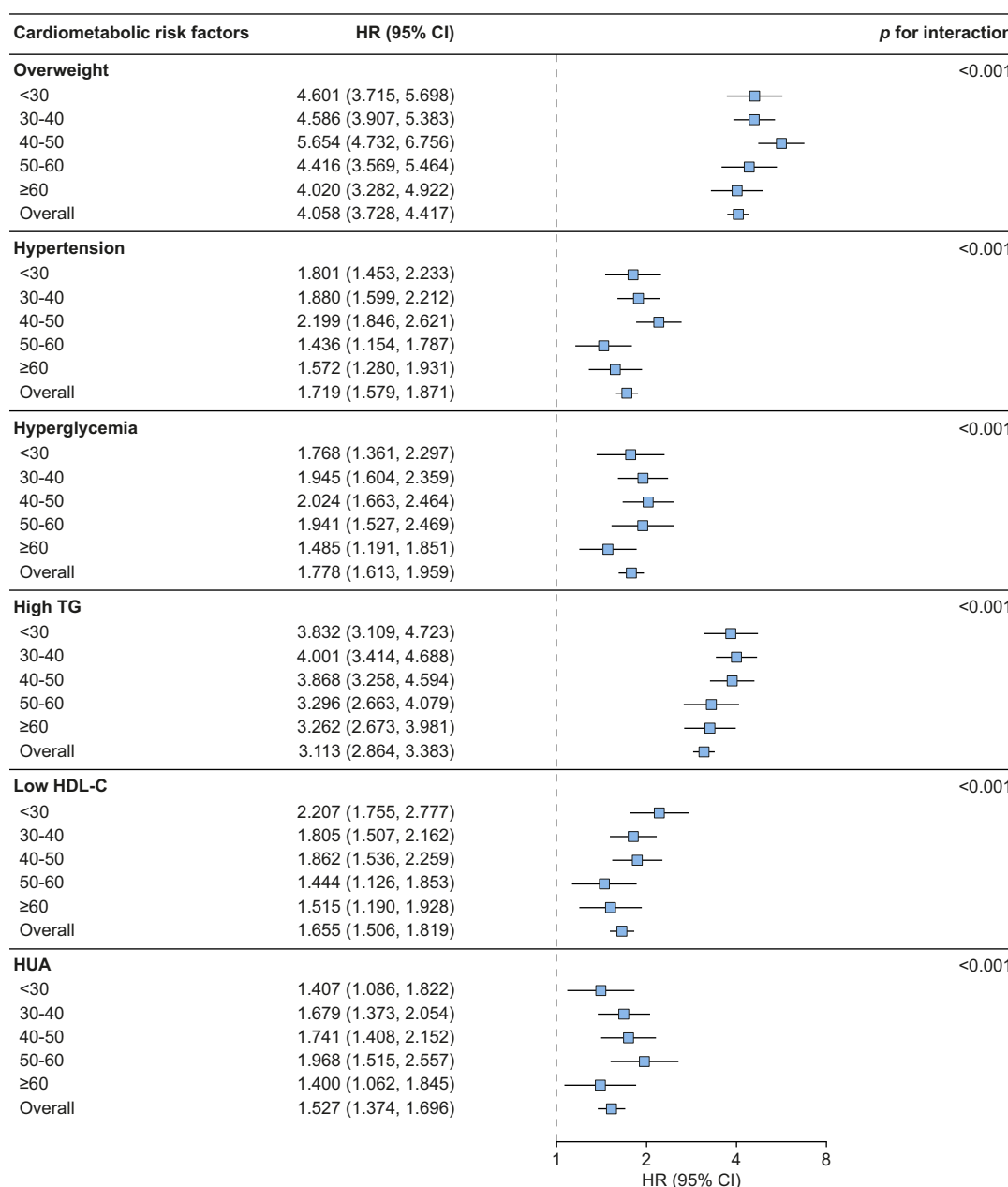


Fig. 1. Relationship between cardiometabolic risk factors and age at onset of NAFLD. All models above adjusted for age, sex, alanine aminotransferase, aspartate aminotransferase, total bilirubin, serum creatinine, blood urea nitrogen. Cox regression analysis was used to determine the association between cardiometabolic risk factors and age at onset of NAFLD. HDL-C, high density lipoprotein cholesterol; HR, hazard ratio; HUA, hyperuricemia; NAFLD, non-alcoholic fatty liver disease; TG, triglyceride.

years.² However, the reasons for the varying age of NAFLD onset remain unclear. A recent shift in terminology from NAFLD to metabolic dysfunction-associated steatotic liver disease (MASLD) has defined five cardiometabolic risk factors strongly associated with NAFLD onset.³ Additionally, previous research has shown that hyperuricemia (HUA) is another cardiometabolic risk factor that increases the likelihood of developing NAFLD.⁴ Therefore, this study aims to investigate the relationship between these six cardiometabolic risk factors and the age of NAFLD onset.

This study analyzed a cohort of 16,152 Chinese non-obese adults who underwent physical examinations and did not have NAFLD at the beginning of the study.⁵ Exclusions were made for individuals with a history of excessive alcohol consumption (>140 g/week for men and >70 g/week for women), viral hepatitis, autoimmune hepatitis, or other known causes of chronic liver disease, as well as those with a BMI ≥ 25 kg/m², LDL cholesterol >3.12 mmol/L, or who were on medication. Liver ultrasound screenings were conducted at each physical examination over an average follow-up period of 2.8 ± 1.1 years, resulting in 2,321 cases of fatty liver. Among these cases, 376 had an onset age of <30 years, 643 were between 30-40 years, 547 between 40-50 years, 355 between 50-60 years, and 400 were 60 years or older. Notably, 43.9% of cases occurred in individuals under 40 years of age, representing close to half of total cases.

All six cardiometabolic risk factors were found to have a significant association with the development of NAFLD at various ages. Across all age groups, individuals with a greater number of cardiometabolic risk factors were found to have a heightened risk of developing NAFLD. Specifically, being overweight and having high levels of triglycerides were highlighted as the two most significant risk factors (Fig. 1 and Table S1). However, the strength of this association varied by age, with a notable shift occurring at 50 years of age. After this age, the association between cardiometabolic risk factors and NAFLD onset tended to decrease, except for low HDL-C and HUA, which showed a different pattern. Notably, low HDL-C was particularly linked to earlier onset of NAFLD, with the highest hazard ratio (HR) observed in individuals under 30 years old (2.207, 95% CI 1.755-2.777) compared to those without NAFLD.

To the best of our knowledge, the current study is the first to examine the relationship between cardiometabolic risk factors

and age of onset of NAFLD. Consistent with previous research on the age of onset of coronary heart disease and diabetes mellitus,^{6,7} our findings support the notion that the association between most cardiometabolic risk factors and the onset of NAFLD is stronger before the age of 50. However, it is important to note that our study was limited to a Chinese population and did not assess changes in cardiometabolic risk factors over time. Therefore, future research should aim to confirm these results in diverse populations and explore how dynamic changes in these factors impact NAFLD.

In summary, the aforementioned cardiometabolic risk factors are linked to a heightened relative risk of developing NAFLD at a younger age, notably low HDL-C. This underscores the importance and advantages of bolstered primary prevention strategies for young adults. Moreover, being overweight carries a significantly higher relative risk compared to other cardiometabolic factors, emphasizing the need to prioritize weight management in NAFLD prevention regardless of age.

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Received 10 April 2024; Received in revised form 18 April 2024;

Accepted 19 April 2024; Available online 3 May 2024

<https://doi.org/10.1016/j.jhep.2024.04.027>

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Financial support

This study was supported by National Natural Science Foundation of China (Grant No. 82270909, No. 81974107).

Conflict of interest

The authors declare no conflicts of interest.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Study concept and design: Tianshu Zeng, Wenkong, and Linfeng He; Statistical analysis and interpretation: Linfeng He; Data acquisition and validation: Wenbin Zheng and Yunfei Liao; All authors wrote and approved final manuscripts.

Data availability statement

The data used in this study are available in the "DATADRYAD" database (www.datadryad.org).

Declaration of Generative AI and AI-assisted technologies in the writing process

During the writing of this work, the authors used Chatgpt 3.5 in order to check grammar, spelling and optimize sentences to enhance the readability of the article. After using this tool/service, the authors reviewed and edited the content as necessary and take full responsibility for the content of the publication.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.04.027>.

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